

Statistics Hybrid Practice Set – S8

Topics

- Population
- Sample
- Mean
- Median
- Mode
- Range
- Variance
- Standard Deviation
- **Quartiles (Q1, Q2, Q3) – New**
- **IQR – New**

Instructions

- Solve on **paper first**.
 - Then solve the **same questions in Excel**.
 - Write the Excel formula you used.
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Q1. Identify Population and Sample

A hospital has **15,000 patients**. A survey is conducted on **450 patients**.

Paper Task

- Population = ?
- Sample = ?

Excel Task

Total Patients	Surveyed Patients
15000	450

Q2. Calculate the Mean

Value
18
22
26
30
34

Paper Task

- Calculate the Mean.

Excel Task

- Write the Excel formula.
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Q3. Find the Median

Value
45
21
33
29
37

Paper Task

- Find the Median.

Excel Task

- Write the Excel formula.
-

Q4. Find the Mode

Value
8
10
10
12
14
16

Paper Task

- Find the Mode.

Excel Task

- Write the Excel formula.
-

Q5. Calculate the Range and Standard Deviation

Value
15
20
25
30
35

Paper Task

Calculate:

- Mean
- Range
- Population Standard Deviation

Excel Task

Write all formulas.

★ New Topic – Quartiles & IQR

Q6. Find Quartiles

Value
5
7
9
11
13
15
17
19
21

Paper Task

Find:

- Q1
- Q2
- Q3

Excel Task

Write the formulas.

Q7. Calculate IQR

Use the same dataset from **Q6**.

Paper Task

Calculate:

- IQR

Excel Task

Write the Excel formula.

Q8. Outlier Detection (IQR Method)

Dataset

Value
12
14
15
16
17
18
19
20
60

Paper Task

1. Find Q1.
2. Find Q3.

3. Find IQR.
4. Calculate:
 - Lower Fence
 - Upper Fence
5. Is **60** an outlier?

Excel Task

Write formulas for:

- Q1
 - Q3
 - IQR
-

Q9. Real Data Analyst Scenario

A sales manager wants to identify **unusually high or unusually low sales values** without letting extreme values affect the analysis.

Which statistical method is the **best** choice?

- A. Mean
- B. Range
- C. IQR (Interquartile Range)
- D. Mode

Explain your answer in one sentence.

Q10. Excel Formula Challenge

Write the Excel formulas for:

1. Mean
2. Median
3. Mode
4. Range
5. Population Variance
6. Population Standard Deviation
7. Q1
8. Q3
9. IQR

Use the range **B2:B10**.